

10

Perturbation theory

It is rarely possible to solve exactly for the dynamics of a system of experimental interest. In these circumstances we use some kind of approximation to tweak the solution to some model system that is as close as possible to the system of interest and yet is simple enough to have analytically solvable dynamics. That is, we treat the difference between the experimental system and the model system as a ‘perturbation’ of the model. Perturbation theory in this sense was an important part of mathematical physics before quantum mechanics appeared on the scene – in fact the development of Hamiltonian mechanics was driven by people who were using perturbation theory to understand the dynamics of the solar system. Interestingly, while perturbation theory in classical mechanics remains an eclectic branch of knowledge that is understood only by a select few, perturbation theory in quantum mechanics is a part of mainstream undergraduate syllabuses. There are two reasons for this. First, analytically soluble models are even rarer in quantum than in classical physics, so more systems have to be modelled approximately. Second, in quantum mechanics perturbation theory is a good deal simpler and works rather better than in classical mechanics.

10.1 Time-independent perturbations

Let H be the Hamiltonian of the experimental system and H_0 the Hamiltonian of the model system for which we have already solved the eigenvalue problem. We hope that $\Delta = H - H_0$ is small and define

$$H_\beta = H_0 + \beta\Delta. \quad (10.1)$$

We can think of H_β as the Hamiltonian of an apparatus that has a knob on it labelled ' β '; when the knob is turned to $\beta = 0$, the apparatus is the model system, and as the knob is turned round to $\beta = 1$, the apparatus is gradually deformed into the system of experimental interest.

We seek the eigenkets $|E\rangle$ and eigenvalues E of H_β as functions of β . Since the Hamiltonian of the apparatus is a continuous function of β , we conjecture that the kets $|E\rangle$ and eigenvalues E are continuous functions of β too. In fact, we conjecture that they are analytic functions¹ of β so they can be expanded as power series

$$|E\rangle = |a\rangle + \beta|b\rangle + \beta^2|c\rangle + \cdots, \quad E = E_a + \beta E_b + \beta^2 E_c + \cdots, \quad (10.2)$$

where $|a\rangle$, $|b\rangle$, etc., are states to be determined and E_a , E_b , etc., are appropriate numbers. When we plug our conjectured forms (10.2) into the eigenvalue equation $H|\psi\rangle = E|\psi\rangle$, we have

$$(H_0 + \beta\Delta)(|a\rangle + \beta|b\rangle + \beta^2|c\rangle + \cdots) = (E_a + \beta E_b + \beta^2 E_c + \cdots)(|a\rangle + \beta|b\rangle + \beta^2|c\rangle + \cdots). \quad (10.3)$$

Since we require the equality to hold for any value of β , we can equate the coefficient of every power of β on either side of the equation:

$$\begin{aligned} \beta^0: & \quad H_0|a\rangle = E_a|a\rangle \\ \beta^1: & \quad H_0|b\rangle + \Delta|a\rangle = E_a|b\rangle + E_b|a\rangle \\ \beta^2: & \quad H_0|c\rangle + \Delta|b\rangle = E_a|c\rangle + E_b|b\rangle + E_c|a\rangle. \end{aligned} \quad (10.4)$$

The first equation simply states that E_a and $|a\rangle$ are an eigenvalue and eigenket of H_0 . Physically, $|a\rangle$ is the state that we will find the system in if we slowly turn the knob back to zero after making a measurement of the energy. Henceforth we shall relabel E_a with E_0 and relabel $|a\rangle$ with $|E_0\rangle$, the zero reminding us of the association with $\beta = 0$ rather than implying that $|E_0\rangle$ is the ground state of the unperturbed system.

To determine E_b we multiply the second equation through by $\langle E_0|$:

$$\langle E_0|H_0|b\rangle + \langle E_0|\Delta|E_0\rangle = E_0\langle E_0|b\rangle + E_b. \quad (10.5)$$

Now from Table 2.1, $\langle E_0|H_0|b\rangle = (\langle b|H_0|E_0\rangle)^* = E_0\langle E_0|b\rangle$. Cancelling this with the identical term on the right, we are left with

$$E_b = \langle E_0|\Delta|E_0\rangle. \quad (10.6)$$

¹ Much interesting physics is associated with phenomena in which a small change in one variable can produce a large change in another (phase changes, narrow resonances, caustics, etc.). In classical physics perturbation theory is bedevilled by such phenomena. In quantum mechanics the conjecture of analytic dependence on β is more successful, but still untrustworthy as we shall see in §10.1.2.

Thus the first-order change in the energy is just the expectation value of the change in the Hamiltonian when the system is in its unperturbed state, which makes good sense intuitively. This is the result that we anticipated in §§8.2.1 and 8.2.2 to estimate the effects on the allowed energies of hydrogen of the spin-orbit and hyperfine Hamiltonians.

Next we determine $|b\rangle$, the first-order change in the state vector. Since the eigenkets $|E_n\rangle$ of H_0 form a complete set of states, we can write $|b\rangle$ as the sum

$$|b\rangle = \sum_k b_k |E_k\rangle. \quad (10.7)$$

In the second of equations (10.4) we replace $|b\rangle$ by this expansion and multiply through by $\langle E_m | \neq \langle E_0 |$ to find

$$b_m = \frac{\langle E_m | \Delta | E_0 \rangle}{E_0 - E_m}. \quad (10.8)$$

This expression determines the coefficient of all kets in equation (10.7) that have energies that differ from the unperturbed value E_0 . For the moment we assume that E_0 is a non-degenerate eigenvalue of H_0 , so there is only one undetermined coefficient, namely that of $|E_0\rangle$. Fortunately we can argue that this coefficient can be taken to be zero from the requirement that $|E\rangle = |E_0\rangle + \beta|b\rangle + O(\beta^2)$ remains correctly normalised. The complete argument is given in Box 10.1 but we can draw a useful analogy with changing a three-dimensional vector so that the condition $|\mathbf{r}| = 1$ is preserved; clearly we have to move $\mathbf{r}$ on the unit sphere and the first-order change in $\mathbf{r}$ is necessarily perpendicular to the original value of $\mathbf{r}$. The quantum-mechanical normalisation condition implies that as β increases $|E\rangle$ moves on a hypersphere in state space and $\langle E_0 | b \rangle = 0$. So we exclude $|E_0\rangle$ from the sum in (10.7) and have that the first-order change to the stationary state is

$$|b\rangle = \sum_{m \neq 0} \frac{\langle E_m | \Delta | E_0 \rangle}{E_0 - E_m} |E_m\rangle. \quad (10.9)$$

According to this relation, $|b\rangle = (d|E\rangle/d\beta)_{\beta=0}$ becomes large when there is a stationary state $|E_m\rangle$ that lies close in energy to the unperturbed state $|E_0\rangle$ and is coupled to the latter in the sense that $\langle E_m | \Delta | E_0 \rangle \neq 0$. This fact will prove crucial in §10.1.2.

To extract the second-order change in E we multiply the third of equations (10.4) by $\langle E_0 |$. Cancelling $\langle E_0 | H_0 | c \rangle$ on $E_0 \langle E_0 | c \rangle$ and using the fact that $\langle E_0 | b \rangle = 0$, we obtain

$$E_c = \langle E_0 | \Delta | b \rangle. \quad (10.10)$$

Substituting for $|b\rangle$ from equation (10.9), we have that the second-order change in E is

$$E_c = \sum_{k \neq 0} \frac{\langle E_0 | \Delta | E_k \rangle \langle E_k | \Delta | E_0 \rangle}{E_0 - E_k} = \sum_{k \neq 0} \frac{|\langle E_0 | \Delta | E_k \rangle|^2}{E_0 - E_k}. \quad (10.11)$$

Box 10.1: Ensuring that $\langle a|b \rangle = 0$

Since the perturbed eigenket should be properly normalised, we have

$$\begin{aligned} 1 &= \langle E|E \rangle = (\langle E_0| + \beta \langle b| + \cdots)(|E_0\rangle + \beta|b\rangle + \cdots) \\ &= 1 + \beta(\langle E_0|b\rangle + \langle b|E_0\rangle) + O(\beta^2). \end{aligned}$$

Equating the coefficient of β on each side of the equation we conclude that $\langle E_0|b\rangle + \langle b|E_0\rangle = 0$, from which it follows that $\langle E_0|b\rangle$ is pure imaginary. The phase of $|E\rangle$ is arbitrary, and we are free to choose this phase independently for each model Hamiltonian H_β . In particular, instead of using $|E\rangle$ we can use $|E'\rangle = e^{i\alpha\beta}|E\rangle$, where α is any real constant: $|E'\rangle$ is our original eigenket but with its phase shifted by a linear function of β . When we expand $|E'\rangle$ in powers of β we have

$$|E'\rangle = |E_0\rangle + \beta|b'\rangle + \cdots,$$

where $|b'\rangle$ is the derivative of $|E'\rangle$ with respect to β evaluated at $\beta = 0$. This is

$$|b'\rangle = \left. \frac{d|E'\rangle}{d\beta} \right|_{\beta=0} = \left. \frac{d}{d\beta} (e^{i\alpha\beta}|E\rangle) \right|_{\beta=0} = i\alpha|E_0\rangle + |b\rangle.$$

Consequently,

$$\langle E_0|b'\rangle = i\alpha + \langle E_0|b\rangle.$$

Since $\langle E_0|b\rangle$ is known to be pure imaginary, it is clear that we can choose α such that $\langle E_0|b'\rangle = 0$. This analysis shows that the phases of the perturbed eigenkets can be chosen such that the first-order perturbation $|b\rangle$ is orthogonal to the unperturbed state $|E_0\rangle$ and one generally assumes that this choice has been made.

10.1.1 Quadratic Stark effect

Let's apply the theory we've developed so far to a hydrogen atom that has been placed in an electric field $\mathbf{E} = -\nabla\Phi$. An externally imposed electric field is small compared to that inside an atom for field strengths up to $\mathcal{E} \simeq 5 \times 10^{11} \text{ V m}^{-1}$ (Problem 8.6) so perturbation theory should yield a good estimate of the shifts in energy level that ordinary fields effect. By the definition of the electrostatic potential Φ , the field changes the energy of the atom by

$$\delta E = e\{\Phi(\mathbf{x}_p) - \Phi(\mathbf{x}_e)\}, \quad (10.12)$$

where $\mathbf{x}_p$ and $\mathbf{x}_e$ are the position vectors of the proton and electron, respectively. We assume that the field changes very little on the scale of the atom, and, as in §8.1, we define $\mathbf{r} = \mathbf{x}_e - \mathbf{x}_p$. Then we may write

$$\delta E \simeq -e\mathbf{r} \cdot \nabla\Phi = e\mathbf{r} \cdot \mathbf{E}. \quad (10.13)$$

We orient our coordinate system so that $\mathbf{E}$ is parallel to the z -axis and use the notation $\mathcal{E} = |\mathbf{E}|$. Then it is clear that the effect of imposing an

external electric field is to add to the unperturbed Hamiltonian a term

$$\Delta = e\mathcal{E}z. \quad (10.14)$$

Suppose the atom is in its ground state $|100\rangle$, where the digits indicate the values of n, l and m . Then from equation (10.6) the first-order energy change in E is

$$E_b = e\mathcal{E}\langle 100|z|100\rangle. \quad (10.15)$$

In §4.1.4 we saw that the expectation value of any component of $\mathbf{x}$ vanishes in a state of well-defined parity. Since the ground-state ket $|100\rangle$ has well-defined (even) parity, $E_b = 0$, and the change in E is dominated by the second-order term E_c . For our perturbation to the ground state of hydrogen, equation (10.11) becomes

$$E_c = e^2\mathcal{E}^2 \sum_{n=2}^{\infty} \sum_{\substack{l \leq n \\ |m| \leq l}} \frac{\langle 100|z|nlm\rangle \langle nlm|z|100\rangle}{E_1 - E_n}. \quad (10.16)$$

Symmetry considerations make it possible to simplify this sum dramatically. First, since $[L_z, z] = 0$ (Table 7.1), $z|nlm\rangle$ is an eigenfunction of L_z with eigenvalue m , and therefore orthogonal to $|100\rangle$ unless $m = 0$. Therefore in equation (10.16) only the terms with $m = 0$ contribute. Second, we can delete from the sum over l all even values of l because, as we saw in §4.1.4, the matrix elements of an odd-parity operator between states of the same parity vanish. In fact, a result proved in Problem 7.21 shows that the terms with $l = 1$ are the only non-vanishing terms in the sums over l in (10.16). Thus

$$E_c = e^2\mathcal{E}^2 \sum_{n=2}^{\infty} \frac{\langle 100|z|n10\rangle \langle n10|z|100\rangle}{E_1 - E_n}. \quad (10.17)$$

It is easy to understand physically why the change in E is proportional to $\mathcal{E}^2$. In response to the external electric field, the probability density of the atom's charge changes by an amount that is proportional to the coefficients b_k , and these coefficients are proportional to $\mathcal{E}$. That is, the field polarises the atom, generating a dipole moment $\mathbf{P}$ that is $\propto \mathcal{E}$. The dipole's energy is $-\mathbf{P} \cdot \mathbf{E}$, so the energy change caused by the field is proportional to $\mathcal{E}^2$.

10.1.2 Linear Stark effect and degenerate perturbation theory

Consider now the shift in the energy of the $n = 2, l = 0$ state of hydrogen when an electric field is applied. The sum over k in (10.16) now includes the term

$$\frac{\langle 200|z|210\rangle \langle 210|z|200\rangle}{E_{20} - E_{21}},$$

which is infinite if we neglect the very small Lamb shift (§8.1.3), because the top is non-zero (Problem 8.12) and the difference of energies on the

bottom vanishes. It hardly seems likely that a negligible field will produce an arbitrarily large change in the energy of the first excited state of hydrogen. So what did we do wrong?

Our error was to assume at the outset that a small stimulus produces a small response as we did when we wrote equations (10.2). Our infinite contribution to E_c can be traced to our expression (10.8) for b_m , which diverges as $E_m \rightarrow E_0$. That is, the change in the wavefunction that a given field produces is inversely proportional to the energy difference between the original state $|E_0\rangle$ and the state $|E_m\rangle$ we are pushing the system towards. This is an entirely reasonable result, analogous to what happens as we push a marble that lies at the bottom of a bowl: the distance the marble moves before coming into equilibrium depends on the curvature of the bowl. In the limit that the curvature goes to zero, and the bottom of the bowl becomes flat, an infinitesimal force will move the marble arbitrarily far, because all locations have the same energy. So we conclude that when the system's initial energy is a degenerate eigenvalue E_d of H_0 , a tiny stimulus is liable to produce a big change in the state (but not the energy) of the system. Disaster will attend an attempt to calculate this abrupt change of state by the approach we have been developing.

So must we just give up in despair? No, because we can see that the only states that are going to acquire a non-negligible amplitude during the abrupt change are ones that have the same energy as E_d . That is, the state to which the system abruptly moves can be expressed as a linear combination of the kets belonging to E_d . In many cases of interest there are only a small number of these (four in the problem of hydrogen on which we are working). What we have to do is to diagonalise the matrix Δ_{ij} formed by Δ squeezed between all pairs of these kets. The eigenkets of Δ in this small subspace will be states of well-defined energy in the slightly perturbed system. As β is ramped up from zero to unity their energies will diverge from E_d . We conjecture that in the instant that β departs from zero, the system's state jumps to the eigenket with the lowest energy, and subsequently stays in this state as β increases. If this conjecture is correct, we should be able to use the perturbation theory we have developed provided we use as basis kets ones that diagonalise Δ as well as H_0 .

So let's diagonalise $e\mathcal{E}z$ in the four-dimensional subspace of hydrogen kets with $n = 2$. When we list the kets in the order $|200\rangle$, $|210\rangle$, $|211\rangle$, $|21-1\rangle$, the matrix of Δ looks like this

$$\Delta_{ij} = e\mathcal{E} \begin{pmatrix} 0 & a & 0 & 0 \\ a^* & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{where } a = \langle 200|z|210\rangle. \quad (10.18)$$

From Problem 8.12 we have that $\langle 200|z|210\rangle = -3a_0$. It is now easy to show that the eigenvalues of Δ are $\pm 3e\mathcal{E}a_0$ and 0, while appropriate eigenkets are $2^{-1/2}(1, \mp 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$. We conclude that as soon as the slightest perturbation is switched on, the system is

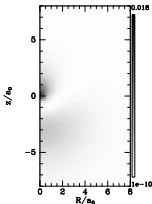


Figure 10.1 The charge distribution of the state $(|200\rangle + |210\rangle)/\sqrt{2}$ is axisymmetric. Here we plot the distribution in the (R, z) plane of cylindrical polar coordinates.

in the state of lowest energy, $|\psi\rangle = 2^{-1/2}(|200\rangle + |210\rangle)$, and we use this state to determine E_δ . We find

$$\begin{aligned} E_\delta &= \frac{1}{2}e\mathcal{E}(\langle 200| + \langle 210|)z(|200\rangle + |210\rangle) \\ &= -3a_0e\mathcal{E}. \end{aligned} \quad (10.19)$$

From our discussion of the quadratic Stark effect, we know that a change in E that is proportional to $\mathcal{E}$ requires the dipole moment $\mathbf{P}$ of an atom to be independent of $\mathcal{E}$. Since E_δ is proportional to $\mathcal{E}$ we conclude that a hydrogen atom in the $n = 2$ state has a permanent electric dipole.

In classical physics this result is to be expected because the orbit of the electron would in general be elliptical, and the time-averaged charge density along the ellipse would be higher at the apocentre than at the pericentre,² because the electron lingers at the apocentre and rushes through the pericentre. Hence the centre of charge would lie on the opposite side of the geometrical centre of the ellipse from the focus at which the proton's cancelling charge lies. Thus, if the electron's orbit were a perfect Kepler ellipse, the atom would have a permanent electric dipole moment parallel to the orbit's major axis. Any deviation of the radial force field from $F \propto r^{-2}$ will cause the major axis of the ellipse to precess, and therefore the time-averaged polarisation of the atom to be zero. In hydrogen the force-field deviates very little from an inverse-square law, so the precession occurs very slowly in the classical picture. Consequently, even a weak external field can prevent precession and thus give rise to a steady electric dipole.

² An orbit's **apocentre** is the point furthest from the attracting body, while the **pericentre** is the point nearest that body.

In the quantum-mechanical picture, shielding shifts the energy of the S state below that of the P states, thus ensuring that, in the absence of an imposed field, the atom is spherical and has no dipole moment. An electric field deprives L^2 of its status as a constant of motion because the field can apply a torque to the atom. Shielding is a very weak effect in hydrogen (because it relies on the vacuum's virtual electrons and positrons), so the S state lies very little below the P states and in even a weak electric field this offset becomes irrelevant. The lowest-energy state becomes $(|200\rangle + |210\rangle)/\sqrt{2}$. This is not an eigenket of L^2 but it is an eigenket of L_z with eigenvalue zero. Thus its angular momentum is perpendicular to the field, as we expect from the classical picture of a Kepler ellipse with its major axis parallel to $\mathbf{E}$. Figure 10.1 shows that in this state the charge distribution comprises a dense cloud around the origin and an extended cloud centred on $R = 0$, $z \simeq -3a_0$. We can think of these clouds as arising from the pericentre and apocentre, respectively, of eccentric orbits that have their major axes roughly aligned with the negative z -axis. The integral $\int d^3\mathbf{x}, z|\psi|^2 = -3a_0$, so in this state the atom has dipole moment $P = +3ea_0$.

10.1.3 Effect of an external magnetic field

When an atom is placed in a magnetic field, the wavelengths of lines in its spectrum change slightly. Much of quantum mechanics emerged from attempts to understand this phenomenon. We now use perturbation theory to explain it.

In Chapter 9 we discussed the motion of a free particle in a uniform magnetic field. Our starting point was the Hamiltonian (9.1), which governs the motion of a free particle of mass m and charge Q in the magnetic field produced by the vector potential $\mathbf{A}$. This is the Hamiltonian of a free particle, $p^2/2m$, with $\mathbf{p}$ replaced by $\mathbf{p} - Q\mathbf{A}$. Hence we can incorporate the effects of a magnetic field on a hydrogen atom by replacing $\mathbf{p}_n$ and $\mathbf{p}_e$ in the gross-structure Hamiltonian (8.1) with $\mathbf{p}_n - e\mathbf{A}$ and $\mathbf{p}_e + e\mathbf{A}$, respectively. With $Z = 1$ the kinetic-energy term in the Hamiltonian then becomes

$$\begin{aligned} H_{KE} &= \frac{(\mathbf{p}_n - e\mathbf{A})^2}{2m_n} + \frac{(\mathbf{p}_e + e\mathbf{A})^2}{2m_e} \\ &= \frac{p_n^2}{2m_n} + \frac{p_e^2}{2m_e} \\ &\quad + \frac{1}{2}e \left\{ \left(\frac{\mathbf{p}_e}{m_e} - \frac{\mathbf{p}_n}{m_n} \right) \cdot \mathbf{A} + \mathbf{A} \cdot \left(\frac{\mathbf{p}_e}{m_e} - \frac{\mathbf{p}_n}{m_n} \right) \right\} + O(A^2). \end{aligned} \quad (10.20)$$

We neglect the terms that are $O(A^2)$ on the grounds that when the field is weak enough for the $O(A)$ terms to be small compared to the terms in the gross-structure Hamiltonian, the $O(A^2)$ terms are negligible.

Equation (8.4) and the corresponding equation for $\partial/\partial\mathbf{x}_n$ imply that

$$\mathbf{p}_e = \frac{m_e}{m_e + m_n} \mathbf{p}_x + \mathbf{p}_r \quad \text{and} \quad \mathbf{p}_n = \frac{m_n}{m_e + m_n} \mathbf{p}_x - \mathbf{p}_r, \quad (10.21)$$

where $\mathbf{p}_\mathbf{X}$ is the momentum associated with the centre-of-mass coordinate $\mathbf{X}$, while $\mathbf{p}_\mathbf{r}$ is the momentum of the reduced particle. From the algebra that leads to equation (8.6a) we know that the first two terms on the right of the second line of equation (10.20) reduce to the kinetic energy of the centre-of-mass motion and of the reduced particle. Using equations (10.21) in the remaining terms on the right of equation (10.20) yields

$$H_{\text{KE}} = \frac{P_{\mathbf{X}}^2}{2(m_e + m_n)} + \frac{P_{\mathbf{r}}^2}{2\mu} + \frac{e}{2\mu}(\mathbf{p}_\mathbf{r} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p}_\mathbf{r}), \quad (10.22)$$

where μ is the mass of the reduced particle (eq. 8.6b). It follows that an external magnetic field adds to the gross-structure Hamiltonian of a hydrogen atom a perturbing Hamiltonian

$$H_B = \frac{e}{2\mu}(\mathbf{p}_\mathbf{r} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p}_\mathbf{r}). \quad (10.23)$$

We are concerned with time-independent fields, and on the scale of the atom the field is likely to be effectively homogeneous, so we may take $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$ (page 255). Then H_B becomes

$$H_B = \frac{e}{4m_e}(\mathbf{p} \cdot \mathbf{B} \times \mathbf{r} + \mathbf{B} \times \mathbf{r} \cdot \mathbf{p}), \quad (10.24)$$

where we have approximated μ by m_e and dropped the subscript on $\mathbf{p}$. The two terms in the bracket on the right can both be transformed into $\mathbf{B} \cdot \mathbf{r} \times \mathbf{p} = \hbar \mathbf{B} \cdot \mathbf{L}$ because (i) these scalar triple products involve only products of *different* components of the three vectors, and (ii) $[x_i, p_j] = 0$ for $i \neq j$. Hence, we do not need to worry about the order of the $\mathbf{r}$ and $\mathbf{p}$ operators and can exploit the usual invariance of a scalar triple product under cyclic interchange of its vectors.

If an atom has more than one unpaired ('valence') electron, each electron will contribute a term of this form to the overall Hamiltonian. We can fold these separate contributions into a single contribution H_B by interpreting $\mathbf{L}$ as the sum of the angular-momentum operators of the individual electrons.

In §8.2.1 we discussed terms that must be added to hydrogen's gross-structure Hamiltonian to account for the effects of the electron's intrinsic dipole moment. We found that the coupling with an external field is generated by the Zeeman spin Hamiltonian (8.76). Adding this to the value of H_B that we have just computed, and orienting our coordinate system so that the z -axis is parallel to $\mathbf{B}$, we arrive at our final result, namely that a uniform magnetic field introduces a perturbation

$$H_{Bs} = \frac{e\hbar}{2m_e} B(L_z + 2S_z) = \mu_B B(J_z + S_z), \quad (10.25)$$

where $\mathbf{S}$ is the sum of the spin operators of all the valence electrons.

The Hamiltonian formed by adding H_{Bs} to the gross-structure Hamiltonian (8.1) commutes with L^2 , L_z , S^2 and S_z . Its eigenkets are simply the

eigenkets of the gross-structure Hamiltonian upgraded to include eigenvalues of S^2 and S_z . The only difference from the situation we studied in §8.1 is that the energies of these eigenkets now depend on both L_z and S_z . Hence, each energy level of the gross-structure Hamiltonian is split by the magnetic field into as many sublevels as $m_l + 2m_s$ can take. For example, if $l = 0$ and $s = \frac{1}{2}$, there are two sublevels, while when $l = 1$ and $s = \frac{1}{2}$ there are five levels in which $m_l + 2m_s$ ranges between ± 2 .

In practice the perturbation H_{Bz} always acts in conjunction with the spin-orbit perturbation H_{SO} of equation (8.77).³ The general case in which H_{Bz} and H_{SO} are comparable, requires numerical solution. The extreme cases in which one operator is larger than the other can be handled analytically.

Paschen-Back effect In a sufficiently strong magnetic field, H_{SO} affects the atom much less than H_{Bz} , so H_{SO} simply perturbs the eigenkets of the Hamiltonian formed by adding H_{Bz} to the gross-structure Hamiltonian. The change in the energy of the state $|n, l, m_l, s, m_s\rangle$ is

$$\begin{aligned} E_b &= \langle n, l, m_l, s, m_s | H_{SO} | n, l, m_l, s, m_s \rangle \\ &= \zeta \langle n, l, m_l, s, m_s | \mathbf{L} \cdot \mathbf{S} | n, l, m_l, s, m_s \rangle, \end{aligned} \quad (10.26)$$

where ζ is a number with dimensions of energy that is independent of m_l and m_s . By writing $\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(L_+ S_- + L_- S_+) + L_z S_z$ (eq. 7.142) we see that $\langle \mathbf{L} \cdot \mathbf{S} \rangle = m_l m_s$. So in a strong magnetic field the eigenenergies are

$$E_{\text{gross}} + \mu_B B(m_l + 2m_s) + \zeta m_l m_s. \quad (10.27)$$

The levels on the extreme right of Figure 10.2 show the energies described by this formula in the case that $l = 1$ and $s = \frac{1}{2}$. The fact that in a strong magnetic field an atom's energies depend on m_l and m_s in this way is known as the **Paschen-Back effect**.

Zeeman effect In a sufficiently weak magnetic field, H_{SO} affects the atom more strongly than H_{Bz} . Then spin-orbit coupling assigns different energies to states that differ in j . Consequently, when we use perturbation theory to calculate the smaller effect of an imposed magnetic field, the degenerate eigenspace in which we have to work is that spanned by the states that have given values of j, l and s but differ in their eigenvalues m of J_z . Fortunately, H_{Bz} is already diagonal within this space because $[J_z, S_z] = 0$. So the shift in the energy of each state is simply

$$E_b = \langle j, m, l, s | H_{Bz} | j, m, l, s \rangle = \mu_B (m + \langle j, m, l, s | S_z | j, m, l, s \rangle). \quad (10.28)$$

As we saw in §7.6, our basis states do not have well-defined values of S_z – in general they are linear combinations of eigenstates of L_z and S_z :

$$|j, m, l, s\rangle = \sum_{m'=-s}^s c_{m'} |l, m - m'\rangle |s, m'\rangle, \quad (10.29)$$

³ There is no spin-orbit coupling for an S state, but an allowed spectral line from an S state will connect to a P state for which there is spin-orbit coupling. Hence the frequencies of allowed transitions inevitably involve spin-orbit coupling.

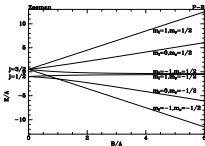


Figure 10.2 Eigenvalues of the spin-dependent Hamiltonian $AL \cdot S + B(L_x + 2S_z)$ as functions of B/A for the case $l = 1$, $s = \frac{1}{2}$. The right side of the diagram (field strong compared to spin-orbit coupling) quantifies the Paschen-Back effect, while the left side of the diagram quantifies the Zeeman effect (weak field). The top and bottom lines on the extreme right show the energies of the states $|1, 1\rangle_+$ and $|1, -1\rangle_-$, which are eigenstates of the full Hamiltonian for all values of B/A .

where the coefficients c_m are Clebsch-Gordan coefficients (eq. 7.152). In any concrete case it is straightforward to calculate the required expectation value of S_z from this expansion. However, a different approach yields a general formula that was important historically.

In the classical picture, spin-orbit coupling causes the vector $\mathbf{S}$ to precess around the invariant vector $\mathbf{J}$. Hence, in this picture the expectation value of $\mathbf{S}$ is equal to the projection of $\mathbf{S}$ onto $\mathbf{J}$.⁴ The classical vector triple product formula enables us to express $\mathbf{S}$ in terms of this projection:

$$\mathbf{J} \times (\mathbf{S} \times \mathbf{J}) = J^2 \mathbf{S} - (\mathbf{S} \cdot \mathbf{J}) \mathbf{J} \quad \text{so} \quad J^2 \mathbf{S} = (\mathbf{S} \cdot \mathbf{J}) \mathbf{J} + \mathbf{J} \times (\mathbf{S} \times \mathbf{L}). \quad (10.30)$$

In the classical picture, the expectation value of the vector triple product on the right side vanishes. If its quantum expectation value were to vanish, the expectation value of the z component of the equation would relate $\langle S_z \rangle$, which we require, to the expectation values of operators that have the states $|j, m, l, s\rangle$ as eigenstates, so our problem would be solved. Motivated by these classical considerations, let's investigate the operator

$$\mathbf{G} = \mathbf{J} \times (\mathbf{S} \times \mathbf{L}) \quad \text{so} \quad G_i = \sum_{jklm} \epsilon_{ijk} J_j \epsilon_{klm} S_l L_m. \quad (10.31)$$

It is straightforward to check that its components commute with the angular-momentum operators J_i in the way we expect the components of a vector to do:

$$[J_i, G_j] = i \sum_k \epsilon_{ijk} G_k. \quad (10.32)$$

⁴ This heuristic argument is often referred to as the **vector model**.

From equation (10.31) it is also evident that $\mathbf{J} \cdot \mathbf{G} = 0$. In Problem 7.21 identical conditions on the operators $\mathbf{L}$ and $\mathbf{x}$ suffice to prove that $\langle x \rangle = 0$ in any state that is an eigenket of L^2 . So the steps of that proof can be retraced with $\mathbf{L}$ replaced by $\mathbf{J}$ and $\mathbf{x}$ replaced by $\mathbf{G}$ to show that for the states of interest $\langle \mathbf{G} \rangle = 0$.

Now that we have established that the quantum-mechanical expectation value of $\mathbf{G}$ does indeed vanish, we reinterpret equation (10.30) as an operator equation, and, from the expectation value of its z component, deduce

$$\langle j, m, l, s | S_z | j, m, l, s \rangle = \frac{(\mathbf{J} \cdot \mathbf{S}) m}{j(j+1)}. \quad (10.33)$$

From equation (8.78) we have

$$\mathbf{J} \cdot \mathbf{S} = \mathbf{L} \cdot \mathbf{S} + S^2 = \frac{1}{2}(J^2 - L^2 + S^2), \quad (10.34)$$

so we find

$$E_B = m g_L \mu_B B \text{ where } g_L = \left(1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)} \right). \quad (10.35)$$

The factor g_L is called the **Landé g factor**. In the early days of quantum theory, when the Bohr atom was taken seriously, people expected the magnetic moment of an electron to be $\pm \mu_B$ and therefore thought a magnetic field would shift energy levels by $\pm \mu_B B$. Equation (10.35) states that the actual shift is $m g_L$ times this. When this factor differed from unity, they spoke of an **anomalous Zeeman effect**.

The left side of Figure 10.2 shows the energy levels described by equation (10.35) in the case $l = 1$, $s = \frac{1}{2}$. The possible values of j are $\frac{3}{2}$ and $\frac{1}{2}$, and the magnetic field splits each of these spin-orbit levels into $2j + 1$ components.

10.2 Variational principle

We now describe a method of estimating energy levels, especially a system's ground-state energy, that does not involve breaking the Hamiltonian down into a part that has known eigenkets and an additional perturbation. In Chapter 11 we shall show that this method yields quite an accurate value for the ionisation energy of helium.

Let H be the Hamiltonian for which we require the eigenvalues E_n and the associated eigenkets $|n\rangle$. We imagine expanding an arbitrary state $|\psi\rangle = \sum_n a_n |n\rangle$ as a linear combination of these eigenkets, and then calculate the expectation value of H in this state as

$$\langle H \rangle = \langle \psi | H | \psi \rangle = \frac{\sum_i |a_i|^2 E_i}{\sum_j |a_j|^2}, \quad (10.36)$$

where we have included the sum of the $|a_j|^2$ on the bottom to cover the possibility that $|\psi\rangle$ is not properly normalised. $\langle H \rangle$ is manifestly

independent of the phase of a_i . We investigate the stationary points of $\langle H \rangle$ with respect to the moduli $|a_i|$ by differentiating equation (10.36) with respect to them:

$$\frac{\partial \langle H \rangle}{\partial |a_k|} = \frac{2|a_k|E_k}{\sum_j |a_j|^2} - \frac{2|a_k| \sum_i |a_i|^2 E_i}{\left(\sum_j |a_j|^2\right)^2}. \quad (10.37)$$

Equating this derivative to zero, we find that the conditions for a stationary point of $\langle H \rangle$ are

$$0 = |a_k| \left(E_k \sum_i |a_i|^2 - \sum_i |a_i|^2 E_i \right) \quad (k = 0, 1, \dots). \quad (10.38)$$

These equations are trivially solved by setting $a_k = 0$ for every k , but then $|\psi\rangle = 0$ so the solution is of no interest. For any value of k for which $a_k \neq 0$, we must have

$$E_k = \frac{\sum_i |a_i|^2 E_i}{\sum_i |a_i|^2}. \quad (10.39)$$

Since the right side of this equation does not depend on k , the equation can be satisfied for at most one value of k , and it clearly is satisfied if we set $a_i = 0$ for $i \neq k$ and $a_k = 1$, so $|\psi\rangle = |k\rangle$. This completes the proof of **Rayleigh's theorem**: *The stationary points of the expectation value of an Hermitian operator occur at the eigenstates of that operator. Moreover, all eigenstates provide stationary points of the operator.* That is, for general $|\psi\rangle$ the number $\langle \psi|H|\psi \rangle$ isn't equal to an eigenvalue of H , but if $\langle \psi|H|\psi \rangle$ is stationary with respect to $|\psi\rangle$ in the sense that it doesn't change when $|\psi\rangle$ is changed by a small amount, Rayleigh's theorem tells us that the number $\langle \psi|H|\psi \rangle$ is an eigenvalue of H . Problem 10.10 gives a geometrical interpretation of Rayleigh's theorem.

The stationary point associated with the ground state is a minimum of $\langle H \rangle$. To see that this is so, we subtract the ground-state energy E_0 from both sides of equation (10.36) and have

$$\langle H \rangle - E_0 = \frac{\sum_i |a_i|^2 (E_i - E_0)}{\sum_j |a_j|^2}. \quad (10.40)$$

Both the top and bottom of the fraction on the right are non-negative, so $\langle H \rangle \geq E_0$. The stationary points of $\langle H \rangle$ associated with excited states are saddle points (Problem 10.15).

The practical use of Rayleigh's theorem is this. We write down a trial wavefunction $\psi_{\mathbf{a}}(\mathbf{x})$ that depends on a number of parameters $a_1, \dots, a_N$. These might, for example, be the coefficients in an expansion of $\psi_{\mathbf{a}}$ as a linear combination of some convenient basis functions $u_i(\mathbf{x})$

$$\psi_{\mathbf{a}}(\mathbf{x}) = \sum_{i=1}^N a_i u_i(\mathbf{x}). \quad (10.41)$$

More often the a_i are parameters in a functional form that is motivated by some physical argument. For example, in Chapter 11 we will treat the variable Z that appears in the hydrogenic wavefunctions of §8.1.2 as one of the a_i . Then we use ψ_a to calculate $\langle H \rangle$ as a function of the a_i and find the stationary points of this function. The minimum value of $\langle H \rangle$ that we obtain in this way clearly provides an upper limit on the ground-state energy E_0 . Moreover, since $\langle H \rangle$ is stationary for the ground-state wavefunction, $\langle H \rangle - E_0$ increases only quadratically in the difference between ψ_a and the ground-state wavefunction ψ_0 . Hence, with even a mediocre fit to ψ_0 this upper limit will lie close to E_0 . This approach to finding eigenvalues and eigenfunctions of the Hamiltonian is called the **variational principle**.

In Problems 10.13 and 10.14 you can explore how the variational principle works in a simple case.

10.3 Time-dependent perturbation theory

We now describe a way of obtaining approximate solutions to the TDSE (2.26) that we shall use to study both scattering of particles and the emission and absorption of radiation by atoms and molecules.

Consider the evolution of a system that is initially in a state that is nearly, but not quite, a stationary state. Specifically, at $t = 0$ it is in the N^{th} eigenstate of a Hamiltonian H_0 that differs by only a small, possibly time-dependent, operator V from the true Hamiltonian H :

$$H = H_0 + V. \quad (10.42)$$

Inspired by (2.32) we expand the solution to the TDSE for this H in the form

$$|\psi\rangle = \sum_n a_n(t) e^{-iE_n t/\hbar} |E_n\rangle, \quad (10.43)$$

where $|E_n\rangle$ is a (time-independent) eigenket of H_0 with eigenvalue E_n . This expansion doesn't restrict $|\psi\rangle$ because $\{|E_n\rangle\}$ is a complete set and the functions $a_n(t)$ are arbitrary. Substituting it into the TDSE we have

$$\begin{aligned} i\hbar \frac{\partial |\psi\rangle}{\partial t} &= (H_0 + V)|\psi\rangle = \sum_n \left(E_n |E_n\rangle + V |E_n\rangle \right) a_n e^{-iE_n t/\hbar} \\ &= \sum_n (i\hbar \dot{a}_n + E_n a_n) e^{-iE_n t/\hbar} |E_n\rangle. \end{aligned} \quad (10.44)$$

We simplify this by multiplying through by $\langle E_k|$:

$$i\hbar \dot{a}_k e^{-iE_k t/\hbar} = \sum_n a_n e^{-iE_n t/\hbar} \langle E_k | V | E_n \rangle. \quad (10.45)$$

This constitutes a set of linear ordinary differential equations for the $a_n(t)$ which must be solved subject to the boundary conditions $a_N(0) = 1$ and

$a_n(0) = 0$ for $n \neq N$. Hence, at the earliest times the term on the right of (10.45) with $n = N$ will dominate the equation of motion of a_k with $k \neq N$, and we have the approximation

$$\dot{a}_k \simeq -\frac{i}{\hbar} e^{-i(E_N - E_k)t/\hbar} \langle E_k | V | E_N \rangle. \quad (10.46)$$

We now assume that any time dependence of V takes the form $V(t) = V_0 e^{i\omega t}$, where V_0 is a time-independent operator. This assumption is in practice not very restrictive because the theory of Fourier analysis enables us to express any operator of the form $V_0 f(t)$, where f is an arbitrary function, as a linear combination of sinusoidally varying operators. Replacing V by $V_0 e^{i\omega t}$ in equation (10.46) and integrating from $t = 0$ we find

$$a_k(t) = \frac{\langle E_k | V_0 | E_N \rangle}{E_N - E_k - \hbar\omega} \left[e^{-i(E_N - E_k - \hbar\omega)t/\hbar} \right]_0^t, \quad (10.47)$$

so the probability that after t the system has made the transition to the k^{th} eigenstate of H_0 is

$$\begin{aligned} P_k(t) &= |a_k|^2 \\ &= \frac{|\langle E_k | V_0 | E_N \rangle|^2}{(E_N - E_k - \hbar\omega)^2} \left\{ 2 - 2 \cos \left(\frac{(E_N - E_k - \hbar\omega)t}{\hbar} \right) \right\} \\ &= 4 |\langle E_k | V_0 | E_N \rangle|^2 \frac{\sin^2((E_N - E_k - \hbar\omega)t/2\hbar)}{(E_N - E_k - \hbar\omega)^2}. \end{aligned} \quad (10.48)$$

For a time of order $\hbar/(E_N - E_k - \hbar\omega)$ this expression grows like t^2 . Subsequently it oscillates.

10.3.1 Fermi golden rule

In many applications of equation (10.48), we need to compute the probability that the system has transitioned to any one of a large number of stationary states of H_0 that we do not distinguish experimentally: our detector will ping if it detects the system in any of these states and we need the probability that our detector will ping. Hence we sum the P_k over all values of k for which the detector will ping.

Let $g(E_k) dE_k$ be the number of stationary states that will make our detector ping and have energies in the interval $(E_k + dE_k, E_k)$. Then with the definition

$$E_{\text{out}} = E_N - \hbar\omega, \quad (10.49)$$

the total pinging probability is

$$\begin{aligned} \sum_k P_k(t) &= 4 \int dE_k g(E_k) |\langle E_k | V | E_N \rangle|^2 \frac{\sin^2((E_{\text{out}} - E_k)t/2\hbar)}{(E_{\text{out}} - E_k)^2} \\ &= \frac{2}{\hbar} \int dx g(E_{\text{out}} - 2\hbar x) |\langle E_{\text{out}} - 2\hbar x | V | E_N \rangle|^2 \frac{\sin^2(xt)}{x^2}, \end{aligned} \quad (10.50)$$

where we've introduced a new variable, $x = (E_{\text{out}} - E_k)/2\hbar$. For given t , the function $f_t(x) \equiv \sin^2(xt)/x^2$ is dominated by a bump around the origin that is of height t^2 and width $2\pi/t$. Hence, the area under the bump is proportional to t and in the limit of large t ,

$$\frac{\sin^2(xt)}{x^2} \propto t\delta(x). \quad (10.51)$$

We find the constant of proportionality by differentiating $\int dx f_t$ with respect to t :

$$\frac{d}{dt} \int_{-\infty}^{\infty} dx \frac{\sin^2(xt)}{x^2} = \int_{-\infty}^{\infty} dx \frac{\sin(2xt)}{x} = \pi, \quad (10.52)$$

where we have used a result, $\int dx \sin x/x = \pi$, from the theory of contour integration in the complex plane. Hence

$$\lim_{t \rightarrow \infty} \frac{\sin^2(xt)}{x^2} = \pi t\delta(x). \quad (10.53)$$

Inserting this relation in (10.50) and integrating over x , we have finally

$$\sum_k P_k = \frac{2\pi t}{\hbar} g(E_{\text{out}}) |\langle \text{out} | V | \text{in} \rangle|^2. \quad (10.54)$$

This equation establishes **Fermi's golden rule**⁵ of perturbation theory: a perturbation $V e^{i\omega t}$ causes a system to transition to a new state lower in energy by $\hbar\omega$ at a rate equal to $2\pi/\hbar$ times the mod-square of the matrix element of V between the initial and final states times the density of relevant states at the final energy. It is easy to see that if the time-dependence of the perturbation were $e^{-i\omega t}$, it would cause transitions at the same rate to states higher in energy by $\hbar\omega$.

10.3.2 Radiative transition rates

We now use equation (10.48) to calculate the rate at which electromagnetic waves induce an atom to make radiative transitions between discrete stationary states. Our treatment is valid when the quantum uncertainty in the electromagnetic field may be neglected, and the field treated as a classical object. This condition is satisfied, for example, in a laser, or at the focus of the antenna of a radio telescope.

Whereas in our derivation of Fermi's golden rule, we took the frequency ω of the perturbation to be fixed and assumed a continuum of final states, now that we are considering the case of a discrete final state, we argue that the electromagnetic field is a superposition of plane waves

⁵ The golden rule was actually first given by P.A.M. Dirac, *Proc. Roy. Soc. A*, **114**, 243 (1927).

of various frequencies, and that we should sum the transition probability (10.48) that each wave independently contributes. Thus we write

$$\sum_{\text{waves}} P_k(t) = 4 \sum_{\text{waves}} |\langle E_k | \delta V_0 | E_N \rangle|^2 \frac{\sin^2((E_N - E_k - \hbar\omega)t/2\hbar)}{(E_N - E_k - \hbar\omega)^2}, \quad (10.55)$$

where δV_0 and ω relate to an individual wave.

In vacuo the electric field of an electromagnetic wave is divergence free, being entirely generated by Faraday's law, $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$. It follows that the whole electromagnetic field of the wave can be described by the vector potential $\mathbf{A}$ through the equations

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \text{and} \quad \mathbf{E} = -\partial \mathbf{A}/\partial t. \quad (10.56)$$

We are considering a superposition of plane waves, which individually have a potential

$$\delta \mathbf{A}(\mathbf{x}, t) = \delta \mathbf{A}_0 \cos(\mathbf{k} \cdot \mathbf{x} - \omega t), \quad (10.57)$$

where $\delta \mathbf{A}_0$ is a constant vector and $\mathbf{k}$ is the wavevector. From equations (10.56) and (10.57) we have that the wave's contribution to the electric field is

$$\delta \mathbf{E}(\mathbf{x}, t) = -\omega \delta \mathbf{A}_0 \sin(\mathbf{k} \cdot \mathbf{x} - \omega t), \quad (10.58)$$

so $\delta \mathbf{E}$ is parallel to $\delta \mathbf{A}_0$. From $\nabla \cdot \delta \mathbf{E} = 0$ it follows that

$$\mathbf{k} \cdot \delta \mathbf{A}_0 = 0, \quad (10.59)$$

so $\mathbf{k} \cdot \delta \mathbf{E} = 0$ and the wave is transverse.

In §10.1.3 we saw that an external electromagnetic field adds to an atom's Hamiltonian the perturbing term (10.23) for each electron. In the present case the perturbation is

$$\delta V(\mathbf{x}, t) = \frac{e}{2m_e} \{ \mathbf{p} \cdot \delta \mathbf{A}_0 \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) + \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) \delta \mathbf{A}_0 \cdot \mathbf{p} \}. \quad (10.60)$$

By virtue of equation (10.59), $\delta \mathbf{A}_0 \cdot \mathbf{p}$ commutes with $\mathbf{k} \cdot \mathbf{x}$ because a component of momentum always commutes with a perpendicular component of position. Since $\delta \mathbf{A}_0$ is a constant, it commutes with $\mathbf{p}$. So we can simplify δV to

$$\begin{aligned} \delta V(\mathbf{x}, t) &= \frac{e}{m_e} \delta \mathbf{A}_0 \cdot \mathbf{p} \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) \\ &= \frac{e}{2m_e} \delta \mathbf{A}_0 \cdot \mathbf{p} \left(e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \right). \end{aligned} \quad (10.61)$$

We now make the approximation that the electromagnetic wavelength is much bigger than the characteristic size of the atom or molecule. This is a good approximation providing the atom or molecule moves between states that are separated in energy by much less than $\alpha m_e c^2$ (Problem 10.21), as will be the case for waves with frequencies that are less

than those of soft X-rays. In this case we will have $\mathbf{k} \cdot \mathbf{x} \ll 1$ for all locations $\mathbf{x}$ in the atom or molecule at which there is significant probability of finding an electron. When this condition is satisfied, it makes sense to expand the factors $e^{\pm i\mathbf{k} \cdot \mathbf{x}}$ in equation (10.61) as a power series and discard all but the constant term. We then have

$$\delta V(\mathbf{x}, t) = \frac{e}{2m_e} \delta \mathbf{A}_0 \cdot \mathbf{p} (e^{-i\omega t} + e^{i\omega t}), \quad (10.62)$$

where we have retained the exponentials in time because large values of t cannot be excluded in the way that we can exclude large values of $\mathbf{x}$. Finally, we note that in the gross-structure Hamiltonian H_0 , $\mathbf{p}$ occurs only in the term $p^2/2m_e$, so $[H_0, \mathbf{x}] = -i(\hbar/m_e)\mathbf{p}$. When we use this relation to eliminate $\mathbf{p}$ from equation (10.62), we have

$$\delta V(\mathbf{x}, t) = i \frac{e\delta A_0}{2\hbar} [H_0, z] (e^{-i\omega t} + e^{i\omega t}), \quad (10.63)$$

where we have chosen to make the z -axis parallel to $\mathbf{A}_0$. Thus a plane electromagnetic wave gives rise to perturbations with both positive and negative frequencies. Above we derived the frequency condition $\omega = (E_N - E_k)/\hbar$ for transitions from $|E_N\rangle$ to $|E_k\rangle$, so the negative frequency perturbation is associated with excitation of the system ($E_k > E_N$), while the positive frequency perturbation is associated with radiative decays.

We identify the time-independent part of δV as the operator δV_0 that occurs in equation (10.55) and then have that the net transition probability is

$$\begin{aligned} \sum_{\text{waves}} P_k(t) &= \frac{e^2}{\hbar^2} \sum_{\text{waves}} (\delta A_0)^2 |\langle E_k | [H_0, z] | E_N \rangle|^2 \frac{\sin^2((E_N - E_k - \hbar\omega)t/2\hbar)}{(E_N - E_k - \hbar\omega)^2} \\ &= \frac{e^2}{4\hbar} (E_k - E_N)^2 |\langle E_k | z | E_N \rangle|^2 \sum_{\text{waves}} (\delta A_0)^2 \frac{\sin^2(xt)}{x^2}, \end{aligned} \quad (10.64a)$$

where

$$x = (E_N - E_k - \hbar\omega)/2\hbar. \quad (10.64b)$$

Even though the expression

$$\rho = \frac{1}{2\mu_0} \{(E/c)^2 + B^2\} \quad (10.65)$$

for the energy density of an electromagnetic field is quadratic in the field amplitudes E and B , the volume-averaged energy density of a superposition of plane waves is just the sum of the energy densities of each individual wave. Moreover, the electric and magnetic energy densities of a plane wave are equal, so the energy density contributed by our plane wave is just twice its electric energy density, and from equations (10.58) and (10.65) we infer that our wave contributes the time-averaged energy density

$$\delta\rho = \frac{\omega^2(\delta A_0)^2}{2\mu_0 c^2} = \frac{1}{2}\omega^2 \epsilon_0 (\delta A_0)^2, \quad (10.66)$$

where the second equality uses $\mu_0 c^2 = 1/\epsilon_0$. Using this expression to eliminate δA_0 from equation (10.64a), we obtain

$$\sum_{\text{waves}} P_k(t) = \frac{e^2}{2\epsilon_0 \hbar^4} (E_k - E_N)^2 |\langle E_k | z | E_N \rangle|^2 \sum_{\text{waves}} \delta \rho \frac{\sin^2(xt)}{\omega^2 x^2}. \quad (10.67)$$

Let $\rho(\omega)$ be the power contained in all waves that have frequencies less than ω . In symbols

$$\rho(\omega) = \sum_{\text{waves with } \omega' < \omega} \delta \rho_{\omega'}. \quad (10.68)$$

Then

$$\sum_{\text{waves}} \delta \rho = \int d\omega \frac{d\rho}{d\omega}. \quad (10.69)$$

When we use this expression to replace the sum on the right side of equation (10.67) by an integral, and we use equation (10.64b) to replace $d\omega$ with $-2dx$, we obtain

$$\sum_{\text{waves}} P_k(t) = \frac{e^2}{\epsilon_0 \hbar^4} (E_k - E_N)^2 |\langle E_k | z | E_N \rangle|^2 \int dx \frac{d\rho}{d\omega} \frac{\sin^2(xt)}{\omega^2 x^2}. \quad (10.70)$$

We now let t become large and exploit equation (10.53) to evaluate the integral. The result is

$$\sum_{\text{waves}} P_k(t) = t \frac{\pi e^2}{\epsilon_0 \hbar^4} |\langle E_k | z | E_N \rangle|^2 \left. \frac{d\rho}{d\omega} \right|_{\omega = (E_N - E_k)/\hbar}. \quad (10.71)$$

The coefficient of t on the right of this equation gives the rate R at which transitions occur. When we express the cumulative energy density of the wave-field in terms of frequency ν rather than angular frequency ω and use equation (8.15) to eliminate ϵ_0 in favour of the Bohr radius, the rate becomes

$$R = \frac{2\pi}{a_0 m_0} |\langle E_k | z | E_N \rangle|^2 \left. \frac{d\rho}{d\nu} \right|_{\nu = |E_N - E_k|/\hbar}. \quad (10.72)$$

When $E_k > E_N$, the negative-frequency term in equation (10.62) gives rise to excitations at an identical rate. Thus we have recovered from a dynamical argument Einstein's famous result that stimulated emission of photons occurs, and that the coefficient B that controls the rate of stimulated emission is equal to the absorption coefficient (Box 10.2). Einstein's prediction of stimulated emission led 38 years later to the demonstration of a maser (§5.2.1) and 44 years later to the construction of the first laser by Theodore Maiman.⁶ In view of this history, it's a remarkable fact that a laser operates in the regime in which the electromagnetic field can

⁶ The word 'laser' is an acronym for 'light amplification by stimulated emission'. Curiously Maiman's paper (*Nature*, **187**, 493 (1960)) about his laser was rejected by the *Physical Review*.

Box 10.2: Einstein A and B coefficients

In 1916, when only the merest fragments of quantum physics were known, Einstein showed (*Verh. Deutsch. Phys. Ges.* **18**, 318) that systems must be capable of both spontaneous and stimulated emission of photons, and that the coefficient of stimulated emission must equal that for absorption of a photon. He obtained these results by requiring that in thermal equilibrium there are equal rates of absorption and emission of photons of a given frequency ν by an ensemble of systems. He considered a frequency ν for which $h\nu = \Delta E$, the energy difference between two states $|1\rangle$ and $|2\rangle$ of the systems. The rate of absorptions he assumed to be $N_{\text{abs}} = B_a N_1 (d\rho/d\nu)$, where B_a is the absorption coefficient, N_1 is the number of systems in the state $|1\rangle$, and $(d\rho/d\nu)$ is the energy density in radiation of frequency ν . The rate of emissions he assumed to be $N_{\text{em}} = B_e N_2 (d\rho/d\nu) + A N_2$, where B_e is the coefficient for induced emission and A is that for spontaneous emission. Equating N_{abs} to N_{em} yields

$$0 = (B_e N_2 - B_a N_1) \frac{d\rho}{d\nu} + A N_2.$$

In thermal equilibrium $N_1 = N_2 e^{h\nu/kT}$ and $d\rho/d\nu$ is given by the Planck function. Using these relations to eliminate N_1 and $d\rho/d\nu$ and then cancelling N_2 , we find

$$0 = (B_e - B_a e^{h\nu/kT}) \frac{8\pi h\nu^3}{c^3 (e^{h\nu/kT} - 1)} - A.$$

In the limit of very large T , $e^{h\nu/kT} \rightarrow 1$, so the factor multiplying the bracket with the B s becomes large, and the contents of this bracket tends to $B_e - B_a$. It follows that these coefficients must be equal. We therefore drop the subscripts on them, take B out of the bracket, cancel the factors with exponentials, and finally deduce that

$$A = 8\pi h(\nu/c)^3 B. \quad (1)$$

be treated as a classical object, as we have done here. Emission of light by a humble candle, by contrast, is an inherently quantum-mechanical phenomenon because it occurs through spontaneous emission. Our treatment does not include spontaneous emission because we have neglected the quantum uncertainty in the electromagnetic field. This uncertainty endows the field with zero-point energy (§3.1), and spontaneous emission can be thought of as emission stimulated by the zero-point energy of the electromagnetic field.

Using the argument given in Box 10.2, Einstein was able to relate the coefficient A of spontaneous emission to B . Einstein's argument does not yield a numerical value for either A or B . Our quantum mechanical treatment has yielded a value for B , and with Einstein's relation (eq. 1 in

Box 10.2) between B and A we can infer the value of A :

$$A = \frac{16\pi^2\hbar\omega^3}{c^3 a_0 m_e} |\langle E_k | z | E_N \rangle|^2. \quad (10.73)$$

From this we can estimate the typical lifetime for radiative decay from an excited state of an atom.

When the radiation density ρ is very small, the number N_2 of atoms in an excited state obeys $\dot{N}_2 = -AN_2$ (Box 10.2), so N_2 decays exponentially with a characteristic time A^{-1} . Unless some symmetry condition causes the matrix element in equation (10.73) to vanish, we expect the value of the matrix element to be $\sim a_0$. So the characteristic radiative lifetime of a state is

$$\tau = A^{-1} = \frac{m_e c^2}{\hbar \nu} \frac{\lambda}{a_0} \frac{1}{16\pi^2 \nu}. \quad (10.74)$$

For an optical transition, $\hbar\nu \sim 2\text{eV}$, $\lambda \sim 650\text{nm} \sim 1.2 \times 10^4 a_0$, and $\nu \sim 4.6 \times 10^{14}\text{Hz}$, so $\tau \sim 4 \times 10^{-8}\text{s}$. It follows that $\sim 10^7$ oscillations of the atom occur before the radiation of energy causes the atom to slump into the lower state.

10.3.3 Selection rules

Equation (10.72) states that the rate of radiative transitions is proportional to the mod-square of the electric dipole operator ez . For this reason the approximation we made, that $\mathbf{k} \cdot \mathbf{x} \ll 1$, is called the **electric dipole approximation**.

There are important circumstances in which symmetry causes the matrix element of the dipole operator to vanish between the initial and final states. Transitions between such states are said to be **forbidden** in contrast to **allowed transitions**, for which the matrix element does not vanish. Some approximations were involved in our derivation of equation (10.72), so the transition rate does not necessarily vanish completely when the matrix element is zero. In fact, forbidden transitions often *do* occur, but at rates that are much smaller than the characteristic rate of allowed transitions (eq. 10.74) because the rate of a forbidden transition is proportional to terms that we could neglect in the derivation of equation (10.72). We now investigate relations between the initial and final states that must be satisfied if the states are to be connected by an allowed transition. Such relations are called **selection rules**. The slower rate of forbidden transitions must be determined by either including the next term of the Taylor expansion of $e^{i\mathbf{k} \cdot \mathbf{x}}$, or taking into account the perturbation $\mu_B \mathbf{S} \cdot \mathbf{B}$ that arises from the interaction of the intrinsic magnetic moment of an electron with the wave's magnetic field.

We are interested in matrix elements between states that are eigenstates of operators that commute with the Hamiltonian H that the atom would have if it were decoupled from electromagnetic waves. The Hamiltonian should include spin-orbit coupling as well as interaction with whatever steady external electric or magnetic fields are being applied. The operator in the matrix element is the component of the position operator

Table 10.1 Selection rules

m	$ m - m' \leq 1$; $m = m'$ for $\mathbf{E}$ parallel to an external field; $ m - m' = 1$ for photon emitted parallel to an external field
j	$ j - j' \leq 1$ but $j = 0 \not\rightarrow j' = 0$
l	$ l - l' = 1$
s	$s = s'$

parallel to the electric field of the radiation that is being either absorbed or emitted.

Even in the presence of an external field, the angular momentum parallel to the field, which we may call J_z , commutes with H , so the kets of interest are labelled with m . Since $[J_z, z] = 0$, the ket $z|E, m\rangle$ is an eigenket of J_z with eigenvalue m . It follows that $\langle E, m|z|E', m'\rangle = 0$ unless $m = m'$. This gives us the first selection rule listed in Table 10.1, namely that when the electric vector of the radiation is parallel to the imposed field, the quantum number m is unchanged by radiation.

If we define $x_{\pm} = x \pm iy$, we have

$$[J_z, x_{\pm}] = iy \pm i(-ix) = \pm x_{\pm}. \quad (10.75)$$

It follows that $x_{\pm}|E, m\rangle$ is an eigenket of J_z with eigenvalue $m \pm 1$, so

$$\begin{aligned} \langle E, m|x|E', m'\rangle &= \frac{1}{2}\langle E, m|(x_+ + x_-)|E', m'\rangle \\ &= 0 \text{ unless } m' = m \pm 1. \end{aligned} \quad (10.76)$$

Obviously the same result applies to the matrix element for y . Hence we have the second selection rule listed in Table 10.1: when the electric vector of the radiation is perpendicular to the imposed field, the quantum number m changes by ± 1 . If the direction of observation is along the imposed field, the electric vector of the radiation must be perpendicular to the field. Hence, in this case m must change by ± 1 . In fact, m increases when a left-hand circularly polarised photon is emitted in the positive z direction, and conversely for the emission of a right-hand polarised photon. When the direction of observation is perpendicular to the imposed field, the electric vector of the radiation can be either perpendicular to the field, in which case m changes by ± 1 , or parallel to the field, and then m does not change.

When there is no imposed field, m may be unchanged or change by ± 1 , and we can observe photons associated with any of these changes in m when observing from any direction.

When there is no imposed field, J^2 commutes with H and the kets of interest are labelled with E , j and m . The selection rule for j can be obtained from the rules for adding angular momentum that were discussed in §7.6: $\langle E, j, m|x_k|E', j', m'\rangle$ vanishes unless it is possible to make a spin- j object by adding a spin-one object to a spin- j' object. For example, the matrix element vanishes if $j = j' = 0$ because spin-one is all you can get by adding a spin-one system to a spin-zero one. Subject to the selection rules on m just discussed, the matrix element does not vanish if $j = 0$

and $j' = 1$, or if $j = 1$ and $j' = 1$, because both a spin-zero system and a spin-one one can be obtained by adding two spin-one subsystems. The matrix element vanishes if $j = 1$ and $j' = 3$ because by adding spin-one and spin-three the smallest spin you can get is spin-two. In summary, the selection rule is $|j - j'| \leq 1$ except that $j = 0 \rightarrow j' = 0$ is forbidden.

The selection rules for j that we have just given follow from a powerful result of group theory, the **Wigner-Eckart theorem**. Unfortunately, a significant amount of group theory is required to prove this theorem. In Appendix J we give a proof of the selection rule for j that builds on the calculation involved in Problem 7.21.

When spin-orbit coupling is weak, the total orbital angular momentum L^2 and the total spin angular momentum S^2 are constants of motion, so their quantum numbers l and s are likely to appear as labels in the kets. Since $[\mathbf{x}, \mathbf{S}] = 0$, it is clear that the selection rule for s is that it should not change. The selection rule for l was derived in Problem 7.21: $|l - l'| = 1$.

Problems

10.1 A harmonic oscillator with mass m and angular frequency ω is perturbed by $\delta H = \epsilon x^2$. (a) What is the exact change in the ground-state energy? Expand this change in powers of ϵ up to order ϵ^2 . (b) Show that the change given by first-order perturbation theory agrees with the exact result to $O(\epsilon)$. (c) Show that the first-order change in the ground state is $|b\rangle = -(\epsilon t^2 / \sqrt{2} \hbar \omega) |E_2\rangle$. (d) Show that second-order perturbation theory yields an energy change $E_c = -\epsilon^2 \hbar / 4 m^2 \omega^3$ in agreement with the exact result.

10.2 The harmonic oscillator of Problem 10.1 is perturbed by $\delta H = \epsilon x$. Show that the perturbed Hamiltonian can be written

$$H = \frac{1}{2m} \left(p^2 + m^2 \omega^2 X^2 - \frac{\epsilon^2}{\omega^2} \right),$$

where $X = x + \epsilon / m \omega^2$ and hence deduce the exact change in the ground-state energy. Interpret these results physically.

What value does first-order perturbation theory give? From perturbation theory determine the coefficient b_1 of the unperturbed first-excited state in the perturbed ground state. Discuss your result in relation to the exact ground state of the perturbed oscillator.

10.3 The harmonic oscillator of Problem 10.1 is perturbed by $\delta H = \epsilon x^4$. Show that the first-order change in the energy of the n^{th} excited state is

$$\delta E = 3(2n^2 + 2n + 1)\epsilon \left(\frac{\hbar}{2m\omega} \right)^2. \quad (10.77)$$

Hint: express x in terms of $A + A^\dagger$.

10.4 The infinite square-well potential $V(x) = 0$ for $|x| < a$ and ∞ for $|x| > a$ is perturbed by the potential $\delta V = cx/a$. Show that to first order in ϵ the energy levels of a particle of mass m are unchanged. Show that even to this order the ground-state wavefunction is changed to

$$\psi_1(x) = \frac{1}{\sqrt{a}} \cos(\pi x/2a) + \frac{16\epsilon}{\pi^2 E_1 \sqrt{a}} \sum_{n=2,4,\dots} (-1)^{n/2} \frac{n}{(n^2-1)^3} \sin(n\pi x/2a),$$

where E_1 is the ground-state energy. Explain physically why this wavefunction does not have well-defined parity but predicts that the particle is more likely to be found on one side of the origin than the other. State with reasons but without further calculation whether the second-order change in the ground-state energy will be positive or negative.

10.5 An atomic nucleus has a finite size, and inside it the electrostatic potential $\Phi(r)$ deviates from $Ze/(4\pi\epsilon_0 r)$. Take the proton's radius to be $a_p \simeq 10^{-15}$ m and its charge density to be uniform. Then treating the difference between Φ and $Ze/(4\pi\epsilon_0 r)$ to be a perturbation on the Hamiltonian of hydrogen, calculate the first-order change in the ground-state energy of hydrogen. Why is the change in the energy of any P state extremely small? Comment on how the magnitude of this energy shift varies with Z in hydrogenic ions of charge Z . Hint: exploit the large difference between a_p and a_0 to approximate the integral you formally require.

10.6 Evaluate the Landé g factor for the case $l = 1$, $s = \frac{1}{2}$ and relate your result to Figure 10.2.

10.7 A particle of mass m moves in the potential $V(x, y) = \frac{1}{2}m\omega^2(x^2 + y^2)$, where ω is a constant. Show that the Hamiltonian can be written as the sum $H_x + H_y$ of the Hamiltonians of two identical one-dimensional harmonic oscillators. Write down the particle's energy spectrum. Write down kets for two stationary states in the first-excited level in terms of the stationary states $|n_x\rangle$ of H_x and $|n_y\rangle$ of H_y . Show that the n^{th} excited level is $n + 1$ fold degenerate.

The oscillator is disturbed by a small potential $H_1 = \lambda xy$. Show that this perturbation lifts the degeneracy of the first excited level, producing states with energies $2\hbar\omega \pm \lambda\hbar/2m\omega$. Give expressions for the corresponding kets.

The mirror operator M is defined such that $\langle x, y|M|\psi\rangle = \langle y, x|\psi\rangle$ for any state $|\psi\rangle$. Explain physically the relationship between the states $|\psi\rangle$ and $M|\psi\rangle$. Show that $[M, H_1] = 0$. Show that $MH_x = H_yM$ and thus that $[M, H] = 0$. What do you infer from these commutation relations?

10.8* The Hamiltonian of a two-state system can be written

$$H = \begin{pmatrix} A_1 + B_1\epsilon & B_2\epsilon \\ B_2\epsilon & A_2 \end{pmatrix}, \quad (10.78)$$

where all quantities are real and ϵ is a small parameter. To first order in ϵ , what are the allowed energies in the cases (a) $A_1 \neq A_2$, and (b) $A_1 = A_2$?

Obtain the exact eigenvalues and recover the results of perturbation theory by expanding in powers of ϵ .

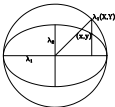


Figure 10.3 The relation of input and output vectors of a 2×2 Hermitian matrix with positive eigenvalues $\lambda_1 > \lambda_2$. An input vector (X, Y) on the unit circle produces the output vector (x, y) that lies on the ellipse that has the eigenvalues as semi-axes.

10.9* For the P states of hydrogen, obtain the shift in energy caused by a weak magnetic field (a) by evaluating the Landé g factor, and (b) by use equation (10.28) and the Clebsch–Gordan coefficients calculated in §7.6.2.

10.10 The 2×2 Hermitian matrix $\mathbf{H}$ has positive eigenvalues $\lambda_1 > \lambda_2$. The vectors (X, Y) and (x, y) are related by

$$\mathbf{H} \cdot \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Show that the points $(\lambda_1 X, \lambda_2 Y)$ and (x, y) are related as shown in Figure 10.3. How does this result generalise to 3×3 matrices? Explain the relation of Rayleigh's theorem to this result.

10.11 We find an upper limit on the ground-state energy of the harmonic oscillator from the trial wavefunction $\psi(x) = (a^2 + x^2)^{-\alpha}$. Using the substitution $x = a \tan \theta$, or otherwise, show that when $\alpha = 1$

$$\int_0^\infty dx |\psi|^2 = \frac{1}{4} \pi a^{-3} \int_0^\infty dx x^2 |\psi|^2 = \frac{1}{4} \pi a^{-1} \int_0^\infty dx |p\psi|^2 = \frac{1}{8} \pi \hbar^2 a^{-5}. \quad (10.79)$$

Hence show that $\langle \psi | H | \psi \rangle / \langle \psi | \psi \rangle$ is minimised by setting $a = 2^{1/4} \ell$, where ℓ is the characteristic length of the oscillator. Show that our upper limit on E_0 is $\hbar\omega/\sqrt{2}$. Plot the final trial wavefunction and the actual ground-state wavefunction and infer how α should be changed to obtain a better trial wavefunction.

10.12* Show that with the trial wavefunction $\psi(x) = (a^2 + x^2)^{-2}$ the variational principle yields an upper limit $E_0 < (\sqrt{7}/5)\hbar\omega \simeq 0.529 \hbar\omega$ on the ground-state energy of the harmonic oscillator.

10.13 Show that for the unnormalised spherically symmetric wavefunction $\psi(r)$ the expectation value of the gross-structure Hamiltonian of hydrogen is

$$\langle H \rangle = \left(\frac{\hbar^2}{2m_e} \int dr r^2 \left| \frac{d\psi}{dr} \right|^2 - \frac{e^2}{4\pi\epsilon_0} \int dr r |\psi|^2 \right) / \int dr r^2 |\psi|^2. \quad (10.80)$$

For the trial wavefunction $\psi_b = e^{-br}$ show that

$$\langle H \rangle = \frac{\hbar^2 b^2}{2m_e} - \frac{e^2 b}{4\pi\epsilon_0},$$

and hence recover the definitions of the Bohr radius and the Rydberg constant.

10.14* Using the result proved in Problem 10.13, show that the trial wavefunction $\psi_b = e^{-b^2 r^2/2}$ yields $-8/(3\pi)\mathcal{R}$ as an estimate of hydrogen's ground-state energy, where $\mathcal{R}$ is the Rydberg constant.

10.15 Show that the stationary point of $\langle \psi | H | \psi \rangle$ associated with an excited state of H is a saddle point. Hint: consider the state $|\psi\rangle = \cos\theta|k\rangle + \sin\theta|l\rangle$, where θ is a parameter.

10.16 At early times ($t \sim -\infty$) a harmonic oscillator of mass m and natural angular frequency ω is in its ground state. A perturbation $\delta H = \mathcal{E}xe^{-t^2/\tau^2}$ is then applied, where $\mathcal{E}$ and τ are constants.

- What is the probability according to first-order theory that by late times the oscillator transitions to its second excited state, $|2\rangle$?
- Show that to first order in δH the probability that the oscillator transitions to the first excited state, $|1\rangle$, is

$$P = \frac{\pi \mathcal{E}^2 \tau^2}{2m\hbar\omega} e^{-\omega^2 \tau^2/2}, \quad (10.81)$$

- Plot P as a function of τ and comment on its behaviour as $\omega\tau \rightarrow 0$ and $\omega\tau \rightarrow \infty$.

10.17 A particle of mass m executes simple harmonic motion at angular frequency ω . Initially it is in its ground state but from $t = 0$ its motion is disturbed by a steady force F . Show that at time $t > 0$ and to first order in F the state is

$$|\psi, t\rangle = e^{-iE_0 t/\hbar}|0\rangle + a_1 e^{-iE_1 t/\hbar}|1\rangle$$

where

$$a_1 = \frac{i}{\sqrt{2m\hbar\omega}} \int_0^t dt' F(t') e^{i\omega t'}.$$

Calculate $\langle x \rangle(t)$ and show that your expression coincides with the classical solution

$$x(t) = \int_0^t dt' F(t') G(t-t'),$$

where the Green's function is $G(t-t') = \sin[\omega(t-t')]/m\omega$. Show that a suitable displacement of the point to which the oscillator's spring is anchored could give rise to the perturbation.

10.18* A particle of mass m is initially trapped by the well with potential $V(x) = -V_0\delta(x)$, where $V_0 > 0$. From $t = 0$ it is disturbed by the time-dependent potential $v(x, t) = -Fxe^{-i\omega t}$. Its subsequent wavefunction can be written

$$|\psi\rangle = a(t)e^{-iE_0 t/\hbar}|0\rangle + \int dk \{b_k(t)|k, e\rangle + c_k(t)|k, o\rangle\} e^{-iE_k t/\hbar}, \quad (10.82)$$

where E_0 is the energy of the bound state $|0\rangle$ and $E_k = \hbar^2 k^2/2m$ and $|k, e\rangle$ and $|k, o\rangle$ are, respectively the even- and odd-parity states of energy E_k (see Problem 5.17). Obtain the equations of motion

$$\begin{aligned} i\hbar \left\{ \dot{a}|0\rangle e^{-iE_0 t/\hbar} + \int dk \left(\dot{b}_k|k, e\rangle + \dot{c}_k|k, o\rangle \right) e^{-iE_k t/\hbar} \right\} \\ = v \left\{ a|0\rangle e^{-iE_0 t/\hbar} + \int dk (b_k|k, e\rangle + c_k|k, o\rangle) e^{-iE_k t/\hbar} \right\}. \end{aligned} \quad (10.83)$$

Given that the free states are normalised such that $\langle k', o|k, o\rangle = \delta(k - k')$, show that to first order in v , $b_k = 0$ for all t , and that

$$c_k(t) = \frac{iF}{\hbar} \langle k, o|x|0\rangle e^{i\Omega_k t/2} \frac{\sin(\Omega_k t/2)}{\Omega_k/2}, \quad \text{where } \Omega_k = \frac{E_k - E_0}{\hbar} - \omega. \quad (10.84)$$

Hence show that at late times the probability that the particle has become free is

$$P_{\text{fr}}(t) = \frac{2\pi m F^2 t}{\hbar^3} \frac{|\langle k, o|x|0\rangle|^2}{k} \Big|_{\Omega_k=0}. \quad (10.85)$$

Given that from Problem 5.17 we have

$$\langle x|0\rangle = \sqrt{K}e^{-K|x|} \quad \text{where} \quad K = \frac{mV_0}{\hbar^2} \quad \text{and} \quad \langle x|k, o\rangle = \frac{1}{\sqrt{\pi}} \sin(kx), \quad (10.86)$$

show that

$$\langle k, o|x|0\rangle = \sqrt{\frac{K}{\pi}} \frac{4kK}{(k^2 + K^2)^2}. \quad (10.87)$$

Hence show that the probability of becoming free is

$$P_{\text{fr}}(t) = \frac{8\hbar F^2 t}{mE_0^2} \frac{\sqrt{E_t/E_0}}{(1 + E_t/E_0)^4}, \quad (10.88)$$

where $E_t > 0$ is the final energy. Check that this expression for P_{fr} is dimensionless and give a physical explanation of the general form of the energy-dependence of $P_{\text{fr}}(t)$.

10.19* A particle travelling with momentum $p = \hbar k > 0$ from $-\infty$ encounters the steep-sided potential well $V(x) = -V_0 < 0$ for $|x| < a$. Use the Fermi golden rule to show that the probability that a particle will be reflected by the well is

$$P_{\text{reflect}} \simeq \frac{V_0^2}{4E^2} \sin^2(2ka),$$

where $E = p^2/2m$. Show that in the limit $E \gg V_0$ this result is consistent with the exact reflection probability derived in Problem 5.10. Hint: adopt periodic boundary conditions so the wavefunctions of the in and out states can be normalised.

10.20* Show that the number of states $g(E) dE d^2\Omega$ with energy in $(E, E + dE)$ and momentum in the solid angle $d^2\Omega$ around $\mathbf{p} = \hbar\mathbf{k}$ of a particle of mass m that moves freely subject to periodic boundary conditions on the walls of a cubical box of side length L is

$$g(E) dE d^2\Omega = \left(\frac{L}{2\pi}\right)^3 \frac{m^{3/2}}{\hbar^3} \sqrt{2E} dE d\Omega^2. \quad (10.89)$$

Hence show from Fermi's golden rule that the cross-section for elastic scattering of such particles by a weak potential $V(\mathbf{x})$ from momentum $\hbar\mathbf{k}$ into the solid angle $d^2\Omega$ around momentum $\hbar\mathbf{k}'$ is

$$d\sigma = \frac{m^2}{(2\pi)^2 \hbar^4} d^2\Omega \left| \int d^3\mathbf{x} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{x}} V(\mathbf{x}) \right|^2. \quad (10.90)$$

Explain in what sense the potential has to be 'weak' for this **Born approximation** to the scattering cross-section to be valid.

10.21 Given that $a_0 = \hbar/(\alpha m_e c)$ show that the product $a_0 k$ of the Bohr radius and the wavenumber of a photon of energy E satisfies

$$a_0 k = \frac{E}{\alpha m_e c^2}. \quad (10.91)$$

Hence show that the wavenumber k_α of an H α photon satisfies $a_0 k_\alpha = \frac{5}{72} \alpha$ and determine λ_α/a_0 . What is the connection between this result and our estimate that $\sim 10^7$ oscillations are required to complete a radiative decay. Does it imply anything about the way the widths of spectral lines from allowed atomic transitions vary with frequency?

10.22 Equation (10.75) implies that $x_\pm$ act as ladder operators for J_z . Why did we not use these operators in §7.1?

10.23 Given that a system's Hamiltonian is of the form

$$H = \frac{p^2}{2m_e} + V(\mathbf{x}) \quad (10.92)$$

show that $[x, [H, x]] = \hbar^2/m_e$. By taking the expectation value of this expression in the state $|k\rangle$, show that

$$\sum_{n \neq k} |\langle n|x|k\rangle|^2 (E_n - E_k) = \frac{\hbar^2}{2m_e}, \quad (10.93)$$

where the sum runs over all the other stationary states.

The **oscillator strength** of a radiative transition $|k\rangle \rightarrow |n\rangle$ is defined to be

$$f_{kn} = \frac{2m_e}{\hbar^2} (E_n - E_k) |\langle n|x|k\rangle|^2. \quad (10.94)$$

Show that $\sum_n f_{kn} = 1$. What is the significance of oscillator strengths for the allowed radiative transition rates of atoms?